

LEONAL ROBIN

Dindigul, Tamil Nadu, India

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SUMMARY

Computer Science and Engineering graduate with strong expertise in Front-end & Back-end web development. Currently working as a Front-end Engineer in an R&D environment at Flam, building scalable web applications and AI-powered interactive media solutions. National-level Hackathon Winner with a proven ability to solve real-world problems through innovative, collaborative engineering.

SKILLS SUMMARY

Programming Languages: Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React.js, Next.js, Node.js, Django, Flask, FastAPI, TensorFlow

Tools & Technologies: Tailwind CSS, MySQL, PostgreSQL, Pinecone, Firebase, AWS, MongoDB, Git

EXPERIENCE

Flam | Front-End Engineer (R&D) Intern

Bangalore, India

NextJs, TypeScript, TailwindCSS, GitHub | Aug 2025 -- Mar 2026

- Built a Remotion-powered video editor with timeline-based sync for text, images, videos, and GLB overlays.
- Engineered a Mastra + OpenAI function-calling pipeline enabling prompt-driven AI control over canvas state, overlay sequencing, and with image-to-video generation so users could produce animated outputs through NLP commands.
- Introduced advanced video editing capabilities including gizmo controls, layer masking, text shaping, and timeline-based keyframe animation with a professional-grade editing experience within a fully browser-based Remotion composition engine.
- Engineered a grouped QR scanning system that bundles multiple related AR experiences under a single scan.
- Built a full-featured AR experience dashboard enabling creation, organization, and publishing of image and ground-tracking experiences—with intuitive editing controls, asset management, and live preview for streamlined content workflows.

MANNIT | SDE Intern

Chennai, India

ReactJs, TypeScript, LLMs, VectorDB, Flask | Feb 2025 -- Jul 2025

- Architected an admin-driven vector database pipeline with automatic category generation and dynamic user-specific data isolation, enabling real-time category reflection that scales with data growth without any hardcoded configuration.
- Resolved Pinecone API quota limits by engineering a hybrid FAISS local indexing layer with in-memory storage, achieving 3--5x faster similarity search and reducing query latency from several seconds down to milliseconds.
- Built PostgreSQL-backed authentication with chat history persistence and admin dashboard for embedding management
- Developed an OAuth-authenticated GitHub app that reads user repositories and delivers AI-powered code reviews covering quality improvements, architectural explanations, refactoring suggestions and auto-generated Mermaid flowcharts
- Engineered scalable codebase analysis for 10,000+ file repos via AST parsing, smart filtering, and batched async processing.

Telecto Energy | SDE Intern

Coimbatore, India

Django, SQL, Node.js, PlotlyJS, MySQL, Jira, REST | Oct 2023 -- May 2024

- Developed and optimized REST APIs using Node.js to ingest and serve large-scale real-time energy datasets with low latency. Normalized and built MySQL schemas to improve query performance and data integrity across energy monitoring pipelines.
- Integrated ML models into the data pipeline to enable predictive analytics on energy consumption patterns; automated model inference on incoming data streams and surfaced actionable insights through interactive PlotlyJS dashboards.
- Built a responsive Django web interface with real-time data visualizations using PlotlyJS, enabling users to monitor live energy metrics, analyze historical usage trends, and make data-driven decisions through an intuitive UI.

PROJECTS

Fake News Detection

Nov 2024 -- Jan 2025

ReactJs, TailwindCSS, Flask, GitHub, AWS, LLM, RAG

- Architected a real-time fact-checking interface integrating OCR pipelines and multimodal AI models (CLIP, GPT-2, Whisper) to analyze uploaded images, text, and video to detect misinformation with 85% accuracy across diverse media formats.
- Extended the detection pipeline to support live and recorded media inputs including YouTube URL, and article URLs—enabling automated misinformation flagging across dynamic web content without requiring manual media uploads. Recognized top25 finalist in Truth Tell Hackathon

SmartClinic

Apr 2026 -- May 2026

Next.js 16, TypeScript, Tailwind CSS, Prisma, PostgreSQL (Supabase), Groq, JWT

- Built a full-stack AI-powered clinic management platform using Next.js 16, Prisma, and Supabase with role-based access for patients, doctors, and admins, featuring appointment scheduling, medical records, and doctor verification workflows.
- Integrated Groq LLM to build an intelligent symptom triage system that classifies urgency, automated visit summary generation from doctor notes, AI-assisted prescription drafting, and semantic natural language search across patient medical records, reducing manual documentation effort for clinicians.

EDUCATION

Amrita Vishwa Vidyapeetham

Coimbatore, India

Bachelor of Technology in Computer Science and Engineering | Sep 2021 -- May 2025

ACHIEVEMENTS AND ACTIVITIES

- NASA Space App Challenge (Oct 2024) — Seismic Data from Apollo and Mars InSight Missions
- ANSAN -- GDSC Project (Jan 2024) — Google Solutions Challenge India Regional Bootcamp
- Pragyan '23 -- National Winner (Mar 2023) — National Level Hackathon at NIT Trichy